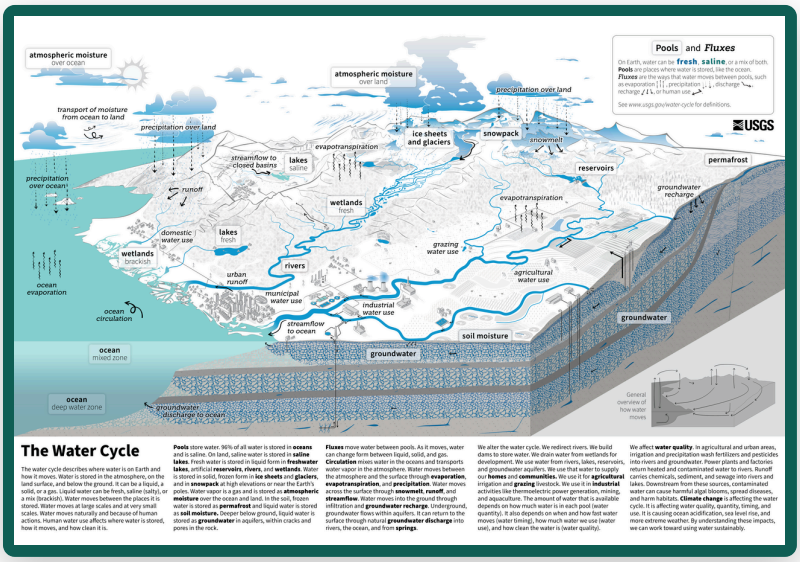
A REGENTS STUDY COMPANION

Earth & Space Science

Everything for the NY State ESS Regents — review, reference & practice



The water cycle — one of Earth's great interconnected systems

Prepared for Vaughn · You've got this! 🌍

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PART I

How to Ace the Exam

The Earth & Space Science Regents is not a memorization marathon — it's a **reading-and-reasoning** test. Most questions come in **clusters**, and many of the answers are hidden right in the passage, graph, or diagram in front of you. If you learn the core science *and* how clusters work, you can figure out questions about topics you've never even heard of. This first section is your game plan.

1.1 How the ESS Regents Works

The exam is built around **cluster questions**. A cluster gives you a **stimulus** — a graph, diagram, data table, map, or short reading passage — and then asks several smaller questions that all refer to that same information. Some questions are **multiple choice**; others are **constructed response**, where you write a short answer in your own words.

Two things make clusters less scary than they look:

- **New vocabulary is usually defined for you.** If a cluster is about "zircons" or "thermohaline circulation," the passage almost always explains the new term — or you can figure it out from context.
- **You get the ESSRT.** The **Earth & Space Sciences Reference Tables** (2024 edition) is a booklet of charts, maps, and diagrams you may use the entire time. Many questions are basically "look it up in the ESSRT." Knowing what's *in* it (see Part III) is one of the biggest, easiest score boosts available.

1.2 Cracking a Cluster — the 7-Step Approach

DO THIS ON EVERY CLUSTER

1. **Read the stimulus carefully first.** For a graph, find the **title, the x-axis, the y-axis, and the units** before you read any question.
2. **Answer each part step by step** — don't jump to conclusions.
3. **Use evidence from the stimulus.** If a question says *"numerical,"* your answer had better include a number.
4. **Check for connections** — a later part may build on an earlier answer.
5. **Find the task word** — what action is the question telling you to do? (See 1.3.)
6. **Spot the "must-haves"** — what does your answer absolutely need to earn the point?
7. **Stuck on a word?** Re-read the stimulus for its definition, then use context clues.

1.3 Task Words — What the Prompt Is Really Asking

Constructed-response questions are graded on whether you did the **specific action** the task word demands. Learn these:

Task word	What to do
Identify	Just name or point out what's asked (e.g., "Identify the fuel for fusion → hydrogen").
Identify a relationship	State how two variables connect ("As CO_2 rises, temperature rises").
Compare	Give similarities/differences using comparison words (more, less, larger, faster).
Describe a property	Give an observable/measurable feature ("igneous rocks have interlocking crystals").
Complete the model / claim	Fill in the missing part of a diagram, sequence, or sentence using science terms.
Construct a claim	Make a conclusion based on evidence.
Use evidence from the graph	Back up your answer with **specific data points or trends** (numbers!).
Analyze data	Find the pattern, trend, or relationship in a table or graph.
Calculate	Do the math / use a formula. Show the number and units.
Explain / Construct an explanation	Give the **cause-and-effect reason** for something ("Hurricanes form because warm ocean water evaporates, rises, and creates low pressure…").
Construct an argument	State a **claim**, support it with **evidence**, and explain your **reasoning**.

Describe a relative amount	Compare quantities using words like more/less, greater/smaller ("the tidal range at A is greater than at B").
Which graph illustrates…	Match a description to the right graph by checking the axes, labels, and the shape of the trend.
Explain the process	Lay out the ordered steps that produce a result (e.g., the steps of weathering).

One golden rule: do exactly what the task word says, and if the prompt mentions **"numerical"** or **"evidence,"** put a specific number or data point in your answer.

1.4 The 7 Crosscutting Concepts

These are the "big ideas" that show up in every science field — graders love when your answers use this kind of thinking:

- **Patterns** — repeated trends help you predict (seasons follow a yearly pattern).
- **Cause & Effect** — identify what causes a change (fossil fuels → more CO_2 → warming).
- **Scale, Proportion & Quantity** — size, time, and amount matter (plate motion is slow per year, huge over millions of years).
- **Systems & System Models** — parts work together; models help predict behavior (the water cycle).
- **Energy & Matter** — track how energy flows and matter cycles (Sun → plants → animals).
- **Structure & Function** — shape determines what something can do.
- **Stability & Change** — what stays the same vs. what changes, and how fast.

1.5 Exam-Day Checklist & Common Traps

EXAM-DAY CHECKLIST — RUN THIS ON EVERY QUESTION

- Read the **title, axes, and units** of every graph *before* the questions.
- **Circle the task word** (identify, explain, calculate, compare, construct…).
- **Underline the numbers** in the stimulus.
- **Check the ESSRT before you guess** — a huge number of answers are right there.
- Answer **with units**, and use **comparative words** (more/less, faster/slower) when comparing.
- For constructed response, write **Claim + Evidence + "because…"** — and include a **number** whenever the prompt says "numerical."

COMMON TRAPS — DON'T MIX THESE UP

- **Weathering** breaks rock *in place*; **erosion** carries it away.
- **Porosity** = space to *hold* water; **permeability** = how easily water *flows through*.
- **Focus** = where a quake starts underground; **epicenter** = the point directly above it on the surface.
- **Rotation** = spin → day & night; **revolution** = orbit → the year.
- **The magnitude scale is backwards:** a *lower* (or negative) absolute magnitude means a *brighter* star.
- **Seasons are caused by Earth's tilt, not its distance** from the Sun.
- **P-waves** travel through solids *and* liquids; **S-waves** travel through solids *only*.
- **Mass extinctions** (sudden, catastrophic) are not the same as ordinary, gradual climate change.

CHAPTER 1

Earth's Systems, Spheres & Feedback

Earth is one giant, connected, ever-changing system. To understand any single event — a hurricane, an ice age, a volcanic eruption — you have to see how Earth's "spheres" interact.

The Four Spheres

- **Geosphere** — the solid Earth: rocks, minerals, landforms, and the interior.
- **Atmosphere** — the layer of gases surrounding Earth; it keeps the planet warm, blocks harmful sunlight, and provides air. The **troposphere** is the lowest layer, where we live and where **weather** happens.
- **Hydrosphere** — all of Earth's water (oceans, rivers, ice, groundwater, water vapor).
- **Biosphere** — all living things and the environments they depend on.

These spheres constantly trade energy and matter. When two spheres shape each other over time, that's **coevolution** — for example, early life produced oxygen that changed the atmosphere, which then allowed new kinds of life to evolve (Chapter 5). The natural events that shape the geosphere — volcanic eruptions, earthquakes, plate movement, weathering, and erosion — are called **geologic processes**, and they're happening all the time, just usually too slowly for us to notice.

Feedback Loops

A **feedback** is when the output of a system loops back to affect the system itself. This is one of the most tested ideas on the whole exam:

- **Positive feedback amplifies change.** Example: melting ice → less white surface to reflect sunlight (lower **albedo**) → more heat absorbed → more warming → *more* melting.
- **Negative feedback reduces change** (it stabilizes). Example: more warming → more evaporation → more clouds → more sunlight reflected → cooling.

Talking About Space and Time

Earth science constantly zooms in and out across both **space** and **time**, and the exam expects you to use these words precisely. The difference between a "range" and a "scale" trips a lot of students up:

- **Spatial range** = the total *area* where something exists or happens (a hurricane's spatial range is all the ocean and land it covers).
- **Spatial scale** = the *size* of the area you're studying — local, regional, or global.
- **Temporal range** = the span of *time* something exists (how long a species lived on Earth).
- **Temporal change** = a change that happens *over time*, whether fast (daily weather) or slow (climate over centuries).

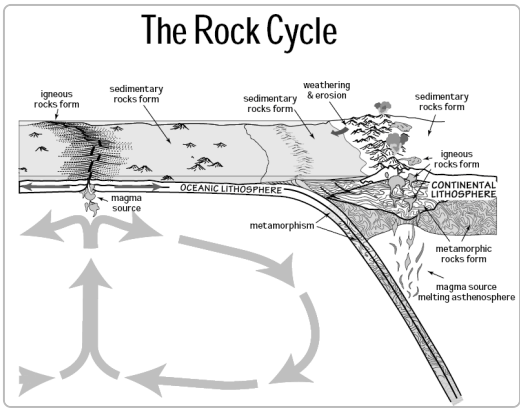
MORE VOCABULARY THE TEST LOVES

Dynamic = constantly changing and active (Earth's climate system is dynamic). **Projected** = predicted for the future based on current data or models (projected sea-level rise for 2100). A **model** is a representation of a system, event, or process — on this test, that's usually an **annotated diagram** you read or complete.

CHAPTER 2

Minerals & the Rock Cycle

Rocks are always being created, destroyed, and recycled. The **rock cycle** has no starting point — and any rock can turn into any other given enough time.



The rock cycle: any rock type can become any other over time.

Minerals — the Building Blocks

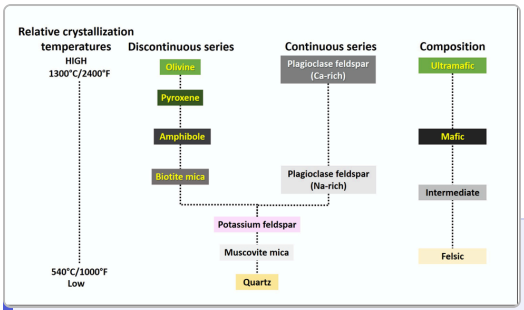
A **mineral** is a substance that is **naturally occurring, inorganic, solid,** with a **definite chemical composition** and a **crystalline structure.** Minerals are the building blocks of rocks. Geologists identify them by physical **properties**: color, **streak** (the color of its powder), **luster** (how it reflects light), **hardness**, **cleavage** (how it breaks), and density. *(Use the Mineral Identification flowchart on ESSRT p.16–17.)*

The Three Rock Types

- **Igneous** — formed from **cooled magma or lava**. If it cools **slowly underground (intrusive/plutonic)**, crystals grow **large**; if it cools **quickly at the surface (extrusive/volcanic)**, the texture is **fine-grained or glassy**. *(This is a classic test point: slow cooling = big crystals.)*
- **Sedimentary** — formed from **compaction and cementation** of sediments. Often **layered** and may contain **fossils** (sandstone, limestone, shale).
- **Metamorphic** — formed when existing rock is changed by **heat and pressure** (slate from shale, marble from limestone). Often **foliated** (banded).

Processes That Drive the Cycle

External forces (at the surface): **weathering & erosion** break rock into sediment; **deposition** lays it down. **Internal forces** (inside Earth): **heat, pressure, melting,** and **cooling**. The full chain: weathering/erosion → deposition → compaction & cementation (sedimentary) → heat & pressure (metamorphic) → melting (magma) → cooling & solidification (igneous) → uplift & exposure → and back to weathering again.



Bowen's Reaction Series (ESSRT p.13) shows the order minerals crystallize as magma cools.

Bowen's Reaction Series predicts the order minerals crystallize from cooling magma — minerals that form at the highest temperatures (like olivine) also weather fastest at the surface.

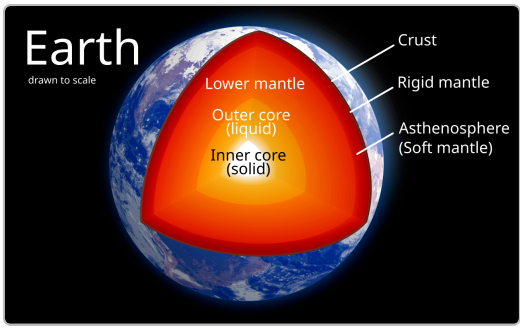
ESSRT PAGES FOR THIS CHAPTER

p.13 Bowen's Reaction Series & Mineral Composition of Igneous Rocks · p.14 Rock Cycle · p.16–17 Mineral Identification · p.8 Bedrock Geology of NY State.

CHAPTER 3

Plate Tectonics, Earthquakes & Volcanoes

Plate tectonics is the master theory of geology — it explains earthquakes, volcanoes, mountains, ocean trenches, and how continents have moved over Earth's history.



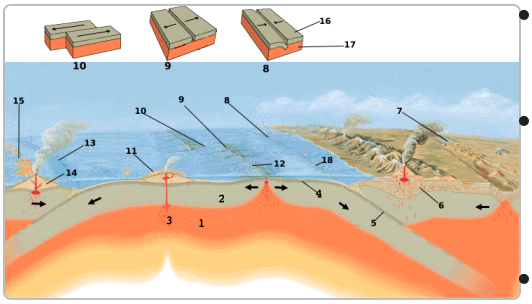
Earth's layers: solid inner core, liquid outer core, solid mantle, thin crust.

Inside the Earth

Evidence from seismic waves, drilling, and magnetism shows Earth has a **solid inner core**, a **liquid outer core**, a **solid (but slowly flowing) mantle**, and a thin **crust**. The **lithosphere** (crust + uppermost mantle) is broken into **tectonic plates**.

Plates move because of **thermal convection** in the mantle: heat from Earth's interior — produced largely by the **radioactive decay** of uranium, thorium, and potassium — makes hot material rise while gravity pulls denser, cooler material down. This slow churning drags the plates along.

Three Types of Plate Boundaries



Divergent, convergent, and transform boundaries.

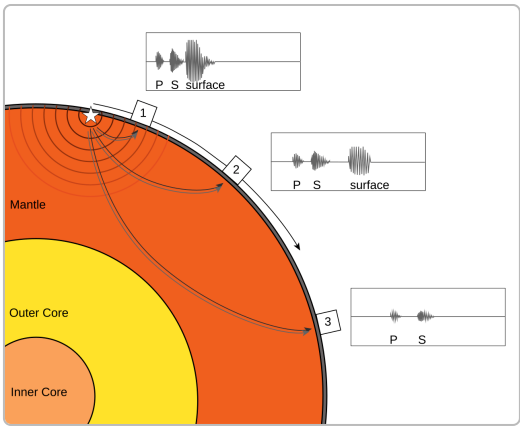
- **Divergent** — plates move **apart**; new crust forms by **sea-floor spreading** (mid-ocean ridges).
- **Convergent** — plates collide; one can subduct under another, building mountains, volcanoes, and deep ocean trenches.
- **Transform** — plates slide **past** each other, causing earthquakes (e.g., the San Andreas Fault).

Hotspots create volcanoes *away* from boundaries, where magma rises through the mantle (Hawaii). The theory is backed by powerful evidence: **paleomagnetism** (magnetic stripes frozen into sea-floor rock), the matching shapes of continents, and **identical fossils found on continents now separated by oceans** — all of which only make sense if the plates once fit together and have since drifted apart.

Earthquakes & Seismic Waves

An earthquake releases energy as **seismic waves**:

- **P-waves (primary) — fastest**; travel through solids, liquids, and gases.
- **S-waves (secondary)** — slower; travel **only through solids**.
- **Surface waves** — travel along the ground; the slowest but they cause the **most damage**. The two kinds are **Love waves** (side-to-side) and **Rayleigh waves** (a rolling, up-and-down motion).



P-waves arrive before S-waves; the gap helps locate the epicenter.

Because P-waves and S-waves travel at different, known speeds, the **gap in their arrival times** at a seismograph station tells scientists how far away the earthquake was — and readings from three stations pinpoint the epicenter.

The **focus** is where the quake starts underground; the **epicenter** is the point on the surface directly above it. **Magnitude** measures energy released; **intensity** measures shaking. **Tsunamis** — giant waves from undersea quakes — travel fast in deep water and grow tall near shore; warning systems and evacuation save lives.

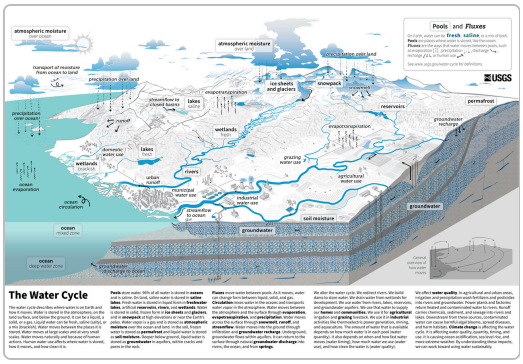
ESSRT PAGES

p.11 Model of Earth's Interior Structure & Cross-Section of Earth's Surface and Interior.

CHAPTER 4

Water & Earth's Changing Surface

Liquid water is what makes Earth *Earth*. Its unusual properties drive weathering, erosion, weather, and life itself.



The water cycle moves water — and energy — through all of Earth's spheres.

Why Water Is Special

Water can **absorb, store, and release large amounts of energy** (which moderates climate), **transmit sunlight**, **expand when it freezes** (so ice floats and cracks rock), **dissolve and transport materials** (the "universal solvent"), and even lower the melting point of rock.

Weathering, Erosion & Deposition

Weathering breaks rock down in place. **Mechanical (physical) weathering** breaks rock without changing it chemically — e.g., **frost wedging** (water freezes in cracks, expands, splits rock) and **abrasion** (grinding by wind, water, or ice). **Chemical weathering** changes the rock's chemistry — **oxidation** (rusting), **hydrolysis**, **carbonation** (weak acid dissolving limestone), and **dissolution**.

Erosion then **transports** the sediment (rivers, glaciers, wind, waves), and **deposition** drops it in new places, building deltas, floodplains, and beaches. Glaciers carve **U-shaped valleys**; rivers carve canyons.

Groundwater

When rain hits the ground it either soaks in (**infiltration**) or flows over the surface (**runoff**). Some is held in soil and plants (**water retention**), and some returns to the air through **evapotranspiration** — the combination of evaporation from surfaces and the water vapor released by plants. The water that soaks in keeps going down until it reaches the **water table**, the boundary that separates the dry **unsaturated zone** above from the fully water-filled **saturated zone** (groundwater) below. Two properties control how groundwater behaves:

- **Porosity** — how much open space (pores) a material has to *hold* water.
- **Permeability** — how easily water can *flow through* those pores.

CHAPTER 5

Earth's History & Dating the Past

Earth is about **4.6 billion years old**. We know this — and the order of events in deep time — thanks to radioactive decay and the rock record.

Radiometric Dating & Half-Life

Unstable atoms break down (**radioactive decay**) at a steady, predictable rate called **exponential decay**. The time for half of a radioactive sample to decay is its **half-life**. Because each element has its own constant half-life, scientists use **radiometric dating** to find the age of rocks and fossils. *(The half-lives are in ESSRT p.15.)* Because decay is so regular, the **same-age oldest minerals** found on Earth, the Moon, and Mars are strong evidence that they all formed together from one disk of material.

How the Solar System & Earth Formed

About 4.6 billion years ago, a giant cloud of gas and dust — a **solar nebula** — collapsed and spun under gravity, forming the hot central **Sun** and a flat, rotating disk. In the disk, particles stuck together by **accretion**, building **planetesimals** that grew into planets and moons. Then came the **Late Heavy Bombardment** (~4.1–3.8 billion years ago), when asteroids and comets cratered the young planets — and may have delivered water and organic material to Earth.

> **WHY WE STUDY THE MOON AND METEORITES**
>
> Active processes on Earth — plate tectonics and erosion by water — have **destroyed most of Earth's oldest rocks**. But the Moon, asteroids, and meteorites have changed little in billions of years, so studying *them* tells us about Earth's earliest history.

Coevolution of the Spheres — How Life Changed the Planet

Life and Earth's systems evolved together — and life literally remade the planet's air. Follow the chain of cause and effect through deep time:

- ~4 bya — life begins as simple **single-celled** organisms in the oceans.

- ~3.5 bya — **cyanobacteria** evolve **photosynthesis**, releasing oxygen as a byproduct. **Stromatolites** (layered mats built by these microbes) are some of the earliest fossil evidence.
- ~2.5–2.0 bya — that oxygen reacts with dissolved iron in the oceans, forming the rusty **Banded Iron Formations**.
- ~2.4 bya — the **Great Oxidation Event**: once the oceans' iron is used up, oxygen finally builds up in the atmosphere, enabling complex, oxygen-using (aerobic) life.
- ~2 bya — **eukaryotic cells** (with a nucleus and organelles) appear.
- ~1 bya — **multicellular** organisms develop specialized cells.
- ~541 mya — the **Cambrian Explosion**: a rapid burst of animal diversity; most modern animal groups appear.
- ~450–360 mya — plants, fungi, and animals **colonize land**, creating new ecosystems.
- ~66 mya — after the last great mass extinction, **mammals and birds** diversify and take over.
- ~6 mya–present — early **hominins** evolve in Africa, leading to modern humans.

Mass extinctions (from volcanic eruptions, asteroid impacts, or climate change) repeatedly wiped out huge numbers of species — and each time, new groups rose to dominate. *(See ESSRT p.6–7, Geologic History of NY State.)*

CHAPTER 6

Earth, Moon & Sun

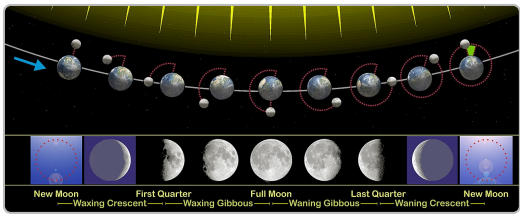
The motions of Earth, the Moon, and the Sun produce day and night, the phases of the Moon, eclipses, the tides, and the seasons — all predictable with a simple model.

Earth's Two Motions

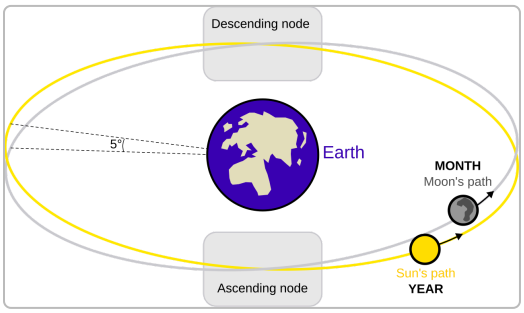
Rotation = Earth spinning on its axis once every ~24 hours → **day and night**, and the Sun's apparent rise in the east and set in the west. **Proofs of rotation: star trails** (circular paths around the celestial pole), the **Foucault pendulum**, and the **Coriolis effect** (winds and currents deflected by Earth's spin). **Revolution** = Earth orbiting the Sun once a year.

Moon Phases & Eclipses

The Moon's **phases** are caused by the changing positions of the Earth, Moon, and Sun — different fractions of the lit half face us as the Moon orbits (**~29.5 days**): New → Waxing Crescent → First Quarter → Waxing Gibbous → Full → Waning Gibbous → Last Quarter → Waning Crescent.



The Moon's phases over its ~29.5-day cycle.



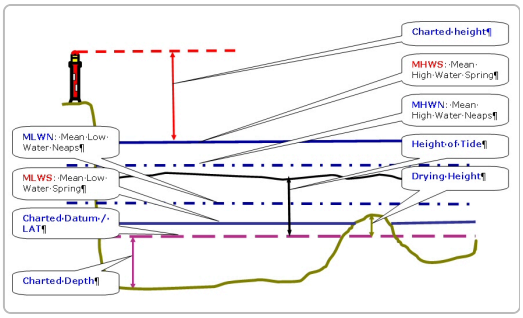
A lunar eclipse: Earth's shadow falls on the Moon (only at full Moon).

Eclipses happen when the three line up: a **solar eclipse** occurs at a **new Moon** (Moon blocks the Sun); a **lunar eclipse** occurs at a **full Moon** (Earth's shadow falls on the Moon).

Tides

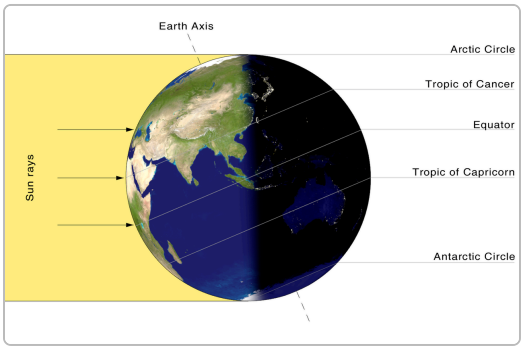
Tides are caused mainly by the **Moon's gravity** (and, less so, the Sun's). **Spring tides** (the highest highs and lowest lows) occur when the Sun, Moon, and Earth are **aligned** — at the

new and full Moon. **Neap tides** (smaller range) occur when the Sun and Moon are at **right angles** — at the first and last quarter.



Spring tides (aligned) vs. neap tides (right angles).

The Seasons



Earth's 23.5° tilt — not its distance from the Sun — causes the seasons.

The seasons are caused by Earth's **23.5° axial tilt**, **not** by its distance from the Sun. As Earth orbits, the hemisphere tilted **toward** the Sun gets **more direct sunlight and longer days** (summer); the hemisphere tilted **away** gets **less direct light and shorter days** (winter). At the equinoxes, neither pole tilts toward the Sun, so day and night are about equal.

CHAPTER 7

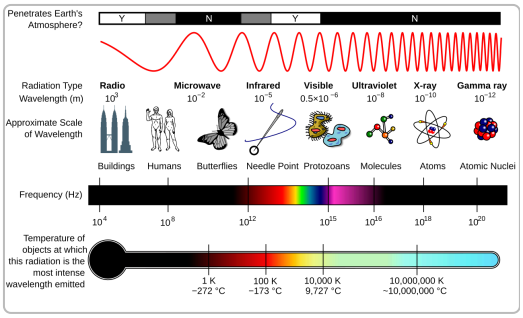
Stars & the Universe

Stars are giant fusion factories, and the light they send us carries a coded message about what they're made of, how hot they are, and how the universe began.

How Stars Make Energy — and Elements

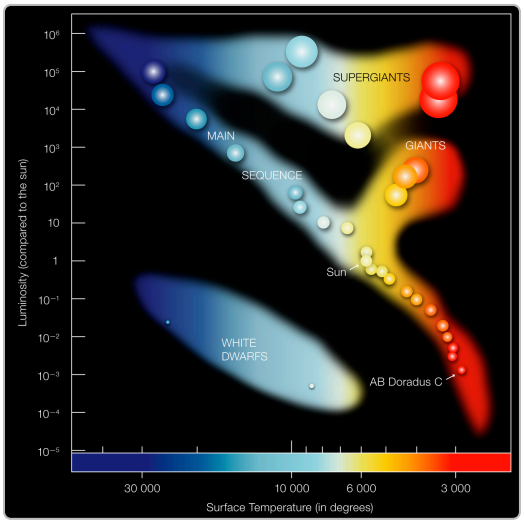
Stars shine by **nuclear fusion**: in the core, hydrogen nuclei fuse into helium, and the tiny bit of **lost mass becomes energy**. Fusion in normal stars builds elements **up to iron**; elements **heavier than iron** are forged when massive stars explode as **supernovae**. A star's whole life is determined by its **initial mass** — and mass also sets its lifespan (bigger stars burn out faster).

Every element emits and absorbs **characteristic wavelengths** of light. By reading a star's **spectrum**, scientists identify which elements it contains, how fast it's moving, and how far away it is.



Each element emits/absorbs specific wavelengths — a fingerprint in starlight.

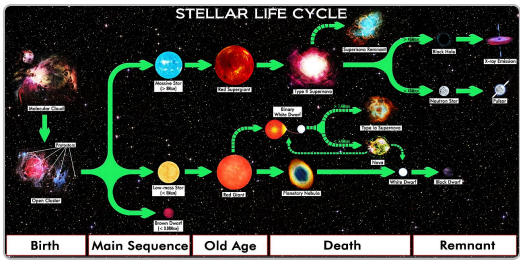
The H-R Diagram & Star Life Cycles



The H-R Diagram (ESSRT p.4) plots temperature vs. brightness.

The **Hertzsprung-Russell (H-R) diagram** plots stars by **surface temperature** (and color) against **brightness**. Four main groups: **main sequence**, **giants**, **supergiants**, and **white dwarfs**. Temperature sets a star's **color** (blue = hot, red = cool). Watch the magnitude trick: a **lower (or negative) absolute magnitude means a brighter star**.

Our **Sun** is an average **main-sequence** star fusing hydrogen into helium — stable for billions of years. Its layers, inside out: **core** (fusion) → **radiative zone** → **convective zone** → **photosphere** (visible surface) → **chromosphere** → **corona**. Low-mass stars end as white dwarfs; high-mass stars explode as supernovae.



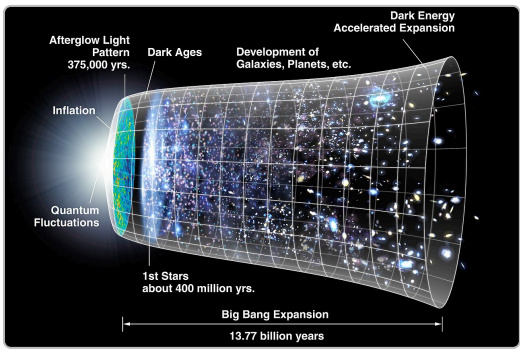
A star's path depends on its mass (ESSRT p.5).

The Solar System: Kepler's Laws

Our solar system is **heliocentric** — planets, asteroids, and comets all orbit the Sun — and it sits within the much larger **Milky Way Galaxy**. To find planets around *other* stars (exoplanets), scientists use the **transit method**: when a planet passes in front of its star, the star's light dips slightly, and that tiny, repeating dip reveals the hidden planet. **Kepler's three laws** describe how anything orbits the Sun: (1) orbits are **ellipses**; (2) a planet moves **faster when closer** to the Sun (perihelion) and slower when farther (aphelion); (3) the farther a planet's average distance, the **longer its period** of revolution. The shape of an orbit is measured by **eccentricity** = distance between foci ÷ length of major axis (a circle = 0).

The Big Bang

The **Big Bang theory** says the universe began ~13.8 billion years ago and has been expanding ever since. The evidence:



The universe has expanded and cooled since the Big Bang.

- **Redshift** — light from distant galaxies is stretched toward longer (redder) wavelengths, and the **farther** the galaxy, the **faster** it's moving away (Hubble's Law). That means the universe is **expanding**.
- **Cosmic Microwave Background** radiation — leftover heat from the early universe.
- The measured **composition** of stars and gas (mostly hydrogen and helium).

ESSRT PAGES

p.2 Nucleosynthesis & Solar System Data · p.3 Emission Spectra & the EM Spectrum · p.4 H-R Diagram · p.5 Life Cycles of Stars.

CHAPTER 8

Weather

Weather is the state of the troposphere right now. Because the atmosphere is so complex, weather can only be predicted **probabilistically** — but the patterns are very learnable.

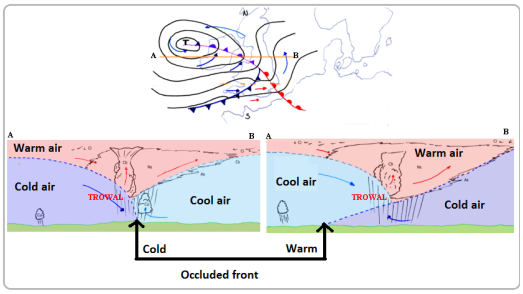
Air Pressure, Humidity & Wind

Air pressure drops when air is **warm**, **humid** (water vapor is lighter than dry air), or at **high altitude**. **Wind blows from high pressure to low pressure**; the steeper the **pressure gradient** (closer the isobars), the **faster** the wind.

Dew point is the temperature at which air becomes saturated and water vapor condenses. The closer the dew point is to the air temperature, the **higher the humidity**; when they're equal, clouds and precipitation are likely.

Air Masses & Fronts

Air masses are huge bodies of air with uniform temperature and moisture; they carry their weather with them. A **front** is the boundary where two air masses meet:



Where two air masses meet, a front forms — and the weather changes.

- **Cold front** — cold air shoves under warm air → **thunderstorms, heavy rain**.
- **Warm front** — warm air slides over cold air → **steady rain or snow**.
- **Stationary front** — neither moves → prolonged clouds and rain.
- **Occluded front** — a cold front overtakes a warm front → complex storms.

Reading Weather Maps & Storms

Station models pack temperature, pressure, wind, cloud cover, and precipitation into one symbol at each location *(see the Weather Map Symbols key, ESSRT p.8)*. **Isobars** connect points of equal pressure — tightly packed isobars mean strong winds. **Hurricanes** form over **warm ocean water**, spin **counterclockwise in the Northern Hemisphere**, and bring wind, rain, and storm surge. **Tornadoes** form from severe thunderstorms with intense,

localized rotating winds. Preparedness — kits, evacuation plans, heeding warnings — saves lives.

ESSRT PAGES

p.8 Weather Map Symbols · p.19 Planetary Wind Belts & Cross-Section of the Lower Atmosphere.

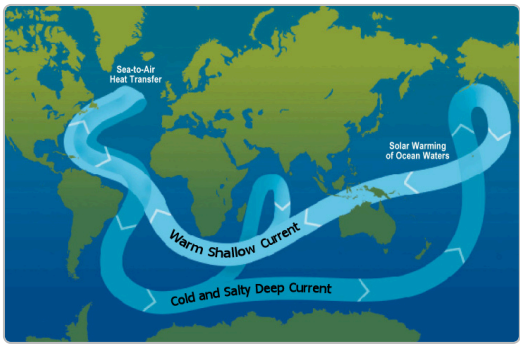
CHAPTER 9

Climate & Human Activity

Weather is what's happening now; **climate** is the long-term average. Climate is shaped by sunlight, oceans, the atmosphere, ice, land, and life — and increasingly by us.

What Controls Climate

Latitude, altitude, and geography all matter, but two big movers are:



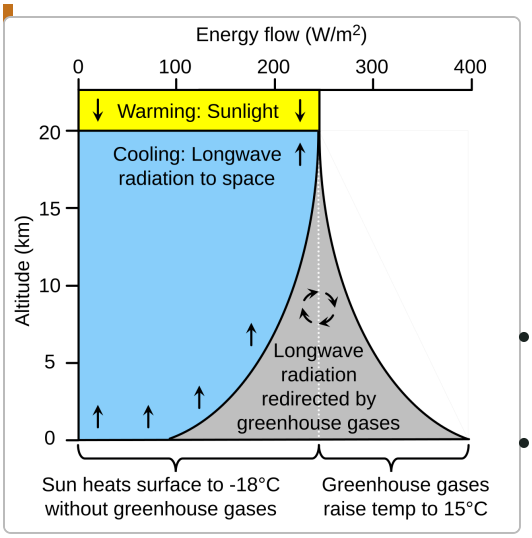
The global "conveyor belt" (thermohaline circulation) moves heat worldwide.

- **Albedo** — how reflective a surface is. Ice and clouds reflect sunlight (high albedo); dark ocean and land absorb it (low albedo).
- **Ocean currents.** Wind-driven **surface currents** like the **Gulf Stream** carry warm water (warming Europe). **Thermohaline circulation (AMOC)** is driven by differences in **temperature and salinity**: cold, salty water sinks in the North Atlantic and pulls warm water north, moving heat around the globe. Melting ice can freshen the water and disrupt it.

Climate changes for many reasons and on many timescales. The geologic record shows it can be driven by interactions among the **Sun's energy output, Earth's orbit and tilt, tectonic events, ocean circulation, volcanic activity, glaciers, vegetation, and human activities**. These play out over wildly different spans — from **sudden** (a volcanic ash cloud cooling things for a year) to **intermediate** (ice ages over tens of thousands of years) to **very long-term** (tectonic cycles over millions of years). The slow, repeating changes in Earth's **orbit shape and axial tilt** — called **Milankovitch cycles** — shift how sunlight is distributed and are a major driver of the ice ages.

Greenhouse Gases & Human Impact

Greenhouse gases (CO_2, methane, water vapor) trap heat in the atmosphere. Human activities — mainly burning **fossil fuels** — have raised CO_2 levels, intensifying the greenhouse effect and warming the planet. **Feedback loops** can amplify this (melting ice → lower albedo → more warming) or dampen it (more clouds → more reflection).



Greenhouse gases trap outgoing heat, warming the surface.

Living Sustainably

Access to resources (minerals, fossil fuels, water) has always shaped human societies, but using them has **economic, social, and environmental costs**. Key human-impact issues:

- **Ocean acidification** — oceans absorb CO_2 and become more acidic, harming coral and shellfish.
- **Saltwater intrusion** — overusing fresh groundwater (or rising seas) lets saltwater contaminate it.
- **Urban heat islands** — pavement and buildings make cities hotter than the countryside.
- **Mining reclamation** — restoring land damaged by mining by reshaping it, replacing soil, and replanting vegetation so ecosystems can recover.
- **Efficient transportation & regulations** — public transit, fuel-efficiency standards, and emissions limits cut pollution and greenhouse gases.

Geology shapes people, too: natural hazards and **geologic events have historically driven human migrations**, and access to resources has guided where and how societies develop. Good solutions weigh **cost, safety, reliability, and environmental and social effects** — and technology plus smart regulations can cut pollution and emissions. Collections of science-based climate strategies, such as **Project Drawdown®**, rank real-world solutions like renewable energy, better food systems, and protecting natural ecosystems by how much greenhouse gas they can keep out of the atmosphere.

PART III

Reference & Practice

Your ESSRT Cheat-Sheet

The Reference Tables are open-book, and NYSED *expects* you to look things up rather than memorize them. So train the reflex: **"When I see ____, I turn to ESSRT page ____."**

When you see a question about…	Turn to ESSRT page
Identifying a mineral	p.16–17 (Mineral ID flowchart)
The rock cycle	p.14
Which minerals crystallize from magma	p.13 (Bowen's Reaction Series)
Earth's interior layers	p.11
Half-lives / radiometric dating math	p.15
Classifying a star (temp & brightness)	p.4 (H-R Diagram)
A star's life cycle	p.5
Planet distances, periods, eccentricities	p.2 (Solar System Data)
Reading weather-map symbols	p.8 (Weather Map Symbols key)
Wind belts / atmosphere structure	p.19
Surface ocean currents	p.20
NY State geologic history / rocks / regions	p.6–10

And here's what lives on each key page:

ESSRT page	What's there / when to use it
p.2	Nucleosynthesis in a massive star; Solar System Data Table (planet distances, periods, eccentricities).
p.3	Emission Spectra of elements; the Electromagnetic Spectrum.
p.4	**H-R Diagram** — classify a star from its temperature & brightness.
p.5	Life Cycles of Stars.
p.6–7	Geologic History of NY State — eras, periods, life, extinctions.
p.8	Bedrock Geology of NY; **Weather Map Symbols** key.

p.9–10	Energy/Mineral Resources & Landscape Regions of NY.
p.11	Earth's Interior Structure.
p.13	Bowen's Reaction Series; Mineral Composition of Igneous Rocks.
p.14	Rock Cycle.
p.15	**Radioactive decay & half-lives** — for radiometric dating problems.
p.16–17	Mineral Identification flowchart.
p.19	Planetary Wind Belts; Lower Atmosphere.
p.20	Surface Ocean Currents.

Glossary of Must-Know Terms

Lithosphere
The rigid outer Earth (crust + uppermost mantle), broken into tectonic plates.

Convection
Heat-driven cycling of material that moves the mantle and plates.

Half-life
The time for half of a radioactive sample to decay; the basis of radiometric dating.

Accretion
The clumping-together of dust and gas that built the planets.

Albedo
How reflective a surface is; high albedo (ice, clouds) reflects sunlight.

Eccentricity
How elliptical an orbit is = distance between foci ÷ length of major axis (circle = 0).

Absolute magnitude
A star's true brightness; a *lower* (or negative) value means a *brighter* star.

Redshift
The stretching of light from receding galaxies — evidence the universe is expanding.

Dew point
The temperature at which air becomes saturated and condensation begins.

Isobars
Map lines of equal air pressure; close together = strong wind.

Feedback
Positive feedback amplifies change (ice-albedo melting); negative feedback stabilizes it.

Porosity vs. permeability

Porosity = space to *hold* water; permeability = ability to *transmit* water.

Coevolution

When spheres (e.g., life and the atmosphere) shape each other's change over time.

Practice Quiz — 40 Questions

Regents-style multiple choice across all topics. Answer key with explanations follows. (Circle your answer.)

1. An igneous rock with very large crystals most likely formed by magma that cooled
   (1) quickly at Earth's surface (2) slowly deep underground (3) instantly in water (4) during weathering
2. Which is **not** a characteristic of a mineral?
   (1) naturally occurring (2) organic (3) crystalline structure (4) definite chemical composition
3. Sedimentary rock is most likely to contain
   (1) large interlocking crystals (2) fossils and layers (3) lava (4) foliated bands
4. The energy that drives mantle convection comes mainly from
   (1) sunlight (2) ocean currents (3) radioactive decay inside Earth (4) tides
5. New oceanic crust forms at a
   (1) convergent boundary (2) transform boundary (3) divergent boundary (4) hotspot island
6. Which seismic wave travels through both solids and liquids and arrives first?
   (1) P-wave (2) S-wave (3) Love wave (4) Rayleigh wave
7. The point on Earth's surface directly above an earthquake's focus is the
   (1) epicenter (2) magnitude (3) fault (4) Moho
8. Earth's layers from outside in are
   (1) crust, mantle, outer core, inner core (2) mantle, crust, core (3) crust, core, mantle (4) core, mantle, crust
9. Frost wedging is an example of
   (1) chemical weathering (2) mechanical weathering (3) deposition (4) erosion by wind
10. The rusting of iron-rich rock is
    (1) abrasion (2) oxidation (3) frost wedging (4) deposition
11. How easily water flows through the pores of a rock is its
    (1) porosity (2) permeability (3) density (4) albedo

12. A property of water that helps it crack rocks is that it

(1) expands when it freezes (2) is a poor solvent (3) blocks all sunlight (4) is denser as a solid than liquid at all temperatures

13. Approximately how old are Earth and the rest of the solar system?

(1) 4.6 million years (2) 13.8 billion years (3) 4.6 billion years (4) 540 million years

14. Scientists determine the absolute age of a rock using

(1) its color (2) radiometric dating (half-lives) (3) its hardness (4) its streak

15. Studying Moon rocks and meteorites helps us learn about early Earth because they

(1) are older than the universe (2) have changed little over billions of years (3) are made of different elements (4) formed after Earth

16. The buildup of oxygen in Earth's early atmosphere was caused mainly by

(1) volcanic eruptions (2) photosynthesis by cyanobacteria (3) asteroid impacts (4) radioactive decay

17. One major cause of mass extinctions has been

(1) large asteroid impacts (2) the Coriolis effect (3) neap tides (4) low albedo

18. The apparent daily rise of the Sun in the east is caused by Earth's

(1) revolution (2) tilt (3) rotation (4) eccentricity

19. A lunar eclipse can occur only during a

(1) new Moon (2) first quarter (3) full Moon (4) waxing crescent

20. Spring tides (largest tidal range) occur when the Sun, Moon, and Earth are

(1) at right angles (2) aligned (new or full Moon) (3) at first quarter (4) farthest apart

21. The seasons are caused primarily by

(1) Earth's changing distance from the Sun (2) the 23.5° tilt of Earth's axis (3) sunspots (4) the Moon's gravity

22. A Foucault pendulum is evidence of Earth's

(1) revolution (2) rotation (3) magnetic field (4) tilt

23. Stars produce energy by

(1) nuclear fission (2) combustion (3) nuclear fusion of hydrogen into helium (4) radioactive decay

24. Elements heavier than iron are formed mainly

(1) in the cores of small stars (2) during supernova explosions (3) on planets (4) by the Big Bang directly

25. What a star will become is determined mainly by its

(1) color (2) distance from Earth (3) initial mass (4) name

26. On an H-R diagram, the Sun is classified as a

(1) white dwarf (2) supergiant (3) main-sequence star (4) red giant

27. A star with a lower (more negative) absolute magnitude is

(1) dimmer (2) brighter (3) cooler (4) closer

28. Redshift of distant galaxies is evidence that the universe is

(1) shrinking (2) staying the same (3) expanding (4) rotating

29. According to Kepler, a planet moves fastest when it is

(1) farthest from the Sun (2) closest to the Sun (3) at its average distance (4) in line with the Moon

30. A nearly circular orbit has an eccentricity closest to

(1) 0.00 (2) 0.50 (3) 0.90 (4) 1.00

31. Which layer of the atmosphere contains our weather?

(1) stratosphere (2) troposphere (3) mesosphere (4) thermosphere

32. Surface winds blow from areas of

(1) low to high pressure (2) high to low pressure (3) warm to cold only (4) land to sea only

33. Adding water vapor to air makes the air pressure

(1) increase (2) decrease (3) stay the same (4) become zero

34. A cold front typically brings

(1) long periods of light drizzle (2) clear, dry skies for days (3) thunderstorms and heavy rain (4) no change in weather

35. On a weather map, closely spaced isobars indicate

(1) calm air (2) strong winds (3) high temperature (4) clear skies

36. In the Northern Hemisphere, hurricanes rotate

(1) clockwise (2) counterclockwise (3) not at all (4) east to west only

37. Melting sea ice lowers Earth's albedo, which leads to more warming. This is an example of

(1) negative feedback (2) positive feedback (3) the Coriolis effect (4) conduction

38. The Gulf Stream is an example of a

(1) wind-driven surface ocean current (2) seismic wave (3) cold front (4) tide

39. The recent rise in atmospheric CO_2 is caused mainly by

(1) volcanic eruptions (2) burning fossil fuels (3) ocean tides (4) the Moon

40. Oceans absorbing CO_2 and becoming more acidic, harming shellfish, is called

(1) salt-water intrusion (2) ocean acidification (3) eutrophication (4) the urban heat island

ANSWER KEY

1. 2 (slow cooling = large crystals) **2.** 2 (minerals are *inorganic*) **3.** 2 **4.** 3 **5.** 3 (sea-floor spreading) **6.** 1 **7.** 1 **8.** 1 **9.** 2 **10.** 2 **11.** 2 **12.** 1 **13.** 3 **14.** 2 **15.** 2 **16.** 2 **17.** 1 **18.** 3 **19.** 3 **20.** 2

21. 2 **22.** 2 **23.** 3 **24.** 2 **25.** 3 **26.** 3 **27.** 2 (lower magnitude = brighter) **28.** 3 **29.** 2 (fastest at perihelion) **30.** 1 (circle = 0) **31.** 2 **32.** 2 **33.** 2 (water vapor is lighter than dry air) **34.** 3 **35.** 2 **36.** 2 **37.** 2 **38.** 1 **39.** 2 **40.** 2

A Worked Cluster Question

Here's how the 7-step approach plays out on a real cluster like the official *"Theia and the Moon"* sample.

THE STIMULUS (SUMMARIZED)

A passage explains the **Giant Impact hypothesis**: a planet-sized body, **Theia**, struck the young Earth; debris accreted into the Moon. It also describes the **Late Heavy Bombardment** (~3.9 bya), when impacts cratered Earth and the Moon. Diagrams show craters and seismic (quake) data for Earth and the Moon.

A typical multiple-choice part: "Which two geologic processes destroyed most of Earth's *earliest* rock record, while 4.2-billion-year-old craters still survive on the Moon?" → **plate tectonic activity and erosion by water.** *(Step 7: the passage and your knowledge tell you Earth is geologically active and wet; the Moon is not.)*

A constructed-response part: "Use radioactive elements and their decay products to support the idea that Moon zircons formed at about the same time as Earth." A full-credit answer names the method and the logic: *"Radiometric dating measures the ratio of a radioactive element to its decay product using its known half-life. Zircons from the Moon and the oldest minerals on Earth give about the same age, so they formed at the same time from the same material." (Step 5–6: the task word is "construct an account," so you must connect evidence — half-lives — to the claim.)*

The takeaway: notice how every answer came from **reading the stimulus + a few core facts** (radiometric dating, Earth is geologically active). That's the whole game.

Mini-Cluster Practice

The real exam is stimulus-based, so here are **three mini-clusters** — each with a graph/table/diagram and a few linked questions, including a written ("constructed-response") part. Read the stimulus first, then answer. Answers follow each cluster.

Mini-Cluster 1 — Planets & Kepler's Laws

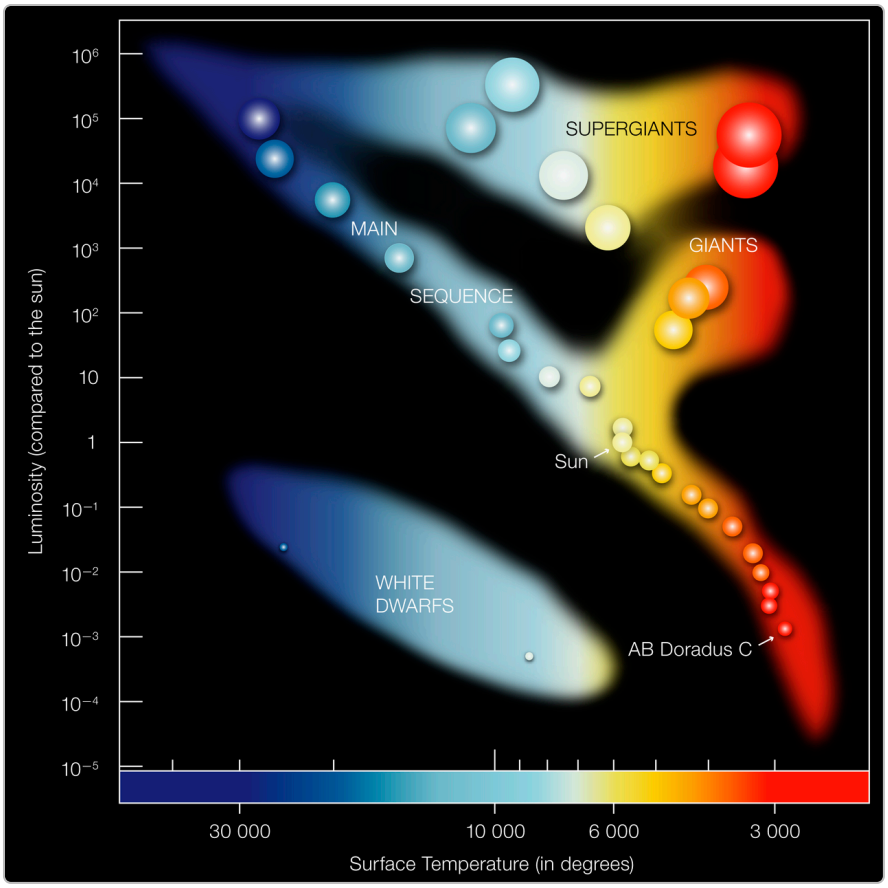
Planet	Avg. distance from Sun (AU)	Period of revolution (years)	Orbital eccentricity
Mercury	0.39	0.24	0.206
Venus	0.72	0.62	0.007
Earth	1.00	1.00	0.017
Mars	1.52	1.88	0.093

1. Using the table, as a planet's distance from the Sun increases, its period of revolution
   (1) increases (2) decreases (3) stays the same (4) varies randomly
2. Which planet has the **most elliptical** (least circular) orbit?
   (1) Mercury (2) Venus (3) Earth (4) Mars
3. Overall, which planet moves **fastest** in its orbit?
   (1) Mercury (2) Venus (3) Earth (4) Mars
4. **(Constructed response)** A new planet is discovered orbiting *beyond* Mars. Predict whether its period of revolution is greater or less than Mars's, and use evidence from the table to support your answer.

ANSWERS — CLUSTER 1

1. (1) increases — Kepler's 3rd law (each row: more distance, longer period). **2.** (1) Mercury (eccentricity 0.206, the highest). **3.** (1) Mercury (closest to the Sun → fastest, Kepler's 2nd law). **4.** *Greater than Mars's.* Evidence: the table shows period always increases with distance (Earth 1.00 AU → 1.00 yr; Mars 1.52 AU → 1.88 yr), so a planet farther than Mars must take **more** than 1.88 years.

Mini-Cluster 2 — Reading the H-R Diagram



Use this H-R Diagram (like ESSRT p.4). Hot, blue stars are on the left; cool, red stars on the right. Brighter stars are higher up.

1. On the H-R diagram, the hottest stars are found on the
   (1) left side (2) right side (3) bottom (4) far corner
2. Our Sun sits along the main sequence. It is classified as a
   (1) white dwarf (2) main-sequence star (3) supergiant (4) red giant
3. Star X has an absolute magnitude of −5; Star Y has +10. Star X is
   (1) dimmer than Y (2) brighter than Y (3) the same brightness (4) cooler than Y
4. **(Constructed response)** Describe how a star's **temperature** and **brightness** change as it leaves the main sequence to become a red giant.

ANSWERS — CLUSTER 2

1. (1) left side (temperature increases to the left). **2.** (2) main-sequence star. **3.** (2) brighter than Y (lower/negative magnitude = brighter). **4.** As it becomes a red giant, the star gets **cooler** (lower surface temperature, shifting right) but **brighter** (greater luminosity / lower magnitude value, shifting up).

Mini-Cluster 3 — CO_2 and Global Temperature

Year	Atmospheric CO_2 (ppm)	Global temperature anomaly (°C)
1900	296	−0.1
1950	311	0.0
2000	369	+0.4
2020	412	+0.9

1. The data show that as atmospheric CO_2 increases, global temperature
   (1) decreases (2) increases (3) stays constant (4) has no relationship
2. The main human activity responsible for the rising CO_2 is
   (1) ocean tides (2) burning fossil fuels (3) volcanic eruptions (4) earthquakes
3. Warmer temperatures melt sea ice, lowering albedo, which causes still more warming. This is
   (1) negative feedback (2) positive feedback (3) the Coriolis effect (4) conduction
4. **(Constructed response)** Use **numerical evidence** from the table to support the claim that CO_2 and global temperature are related.

ANSWERS — CLUSTER 3

1. (2) increases. **2.** (2) burning fossil fuels. **3.** (2) positive feedback. **4.** Full-credit answers use numbers from the table, e.g.: *"From 1900 to 2020, CO_2 rose from 296 ppm to 412 ppm, and over the same time the temperature anomaly rose from −0.1 °C to +0.9 °C. Because both increased together, the data support the claim that they are related."*